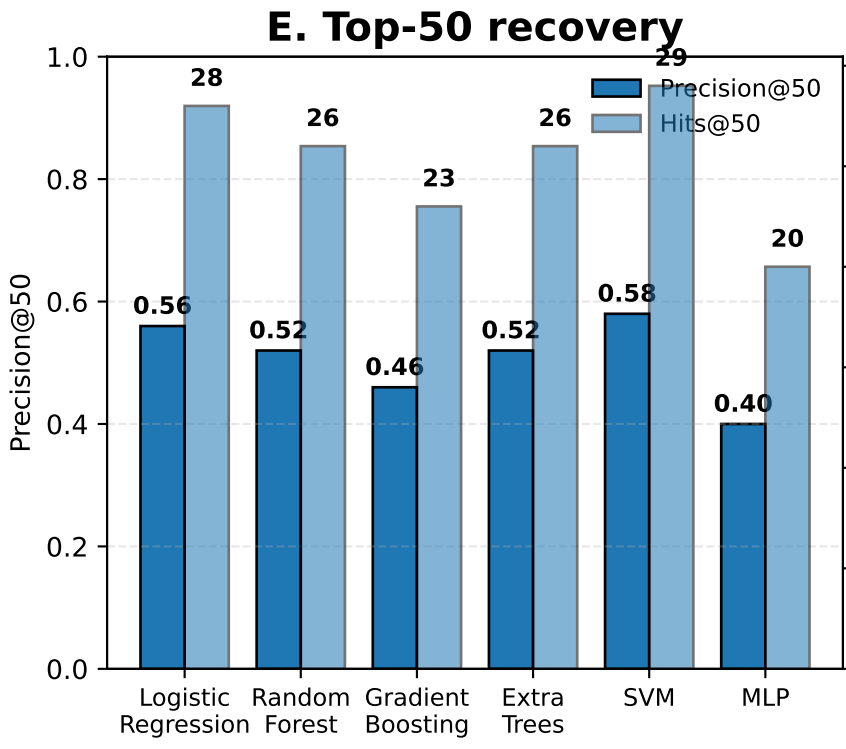
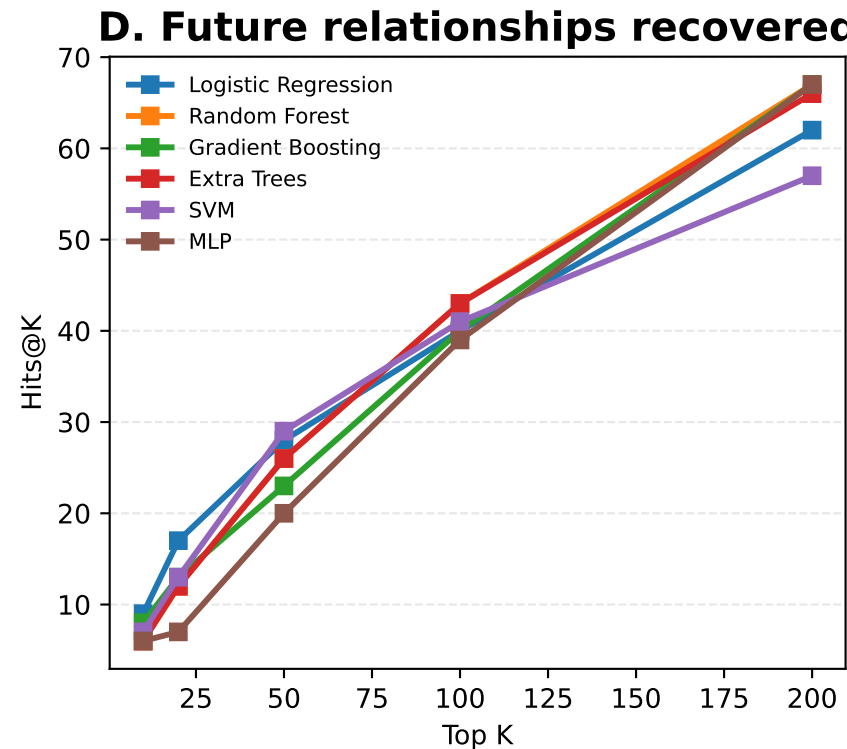
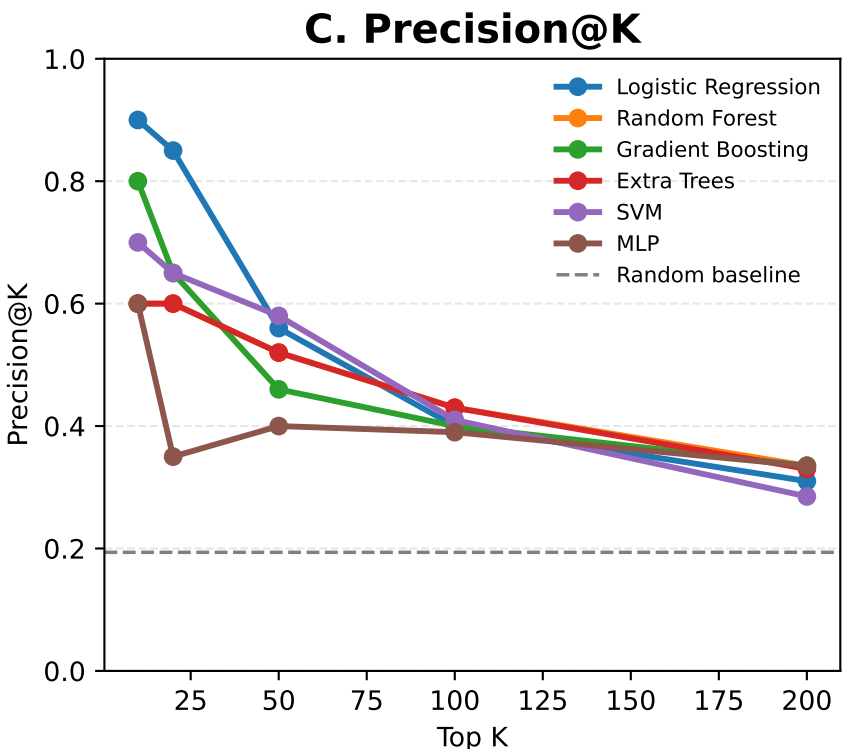
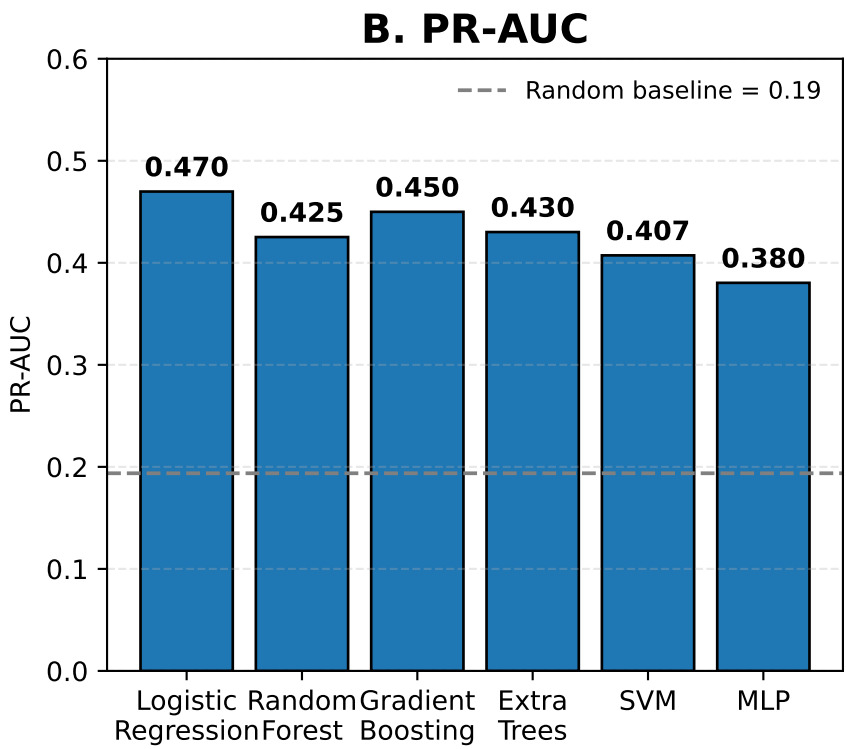
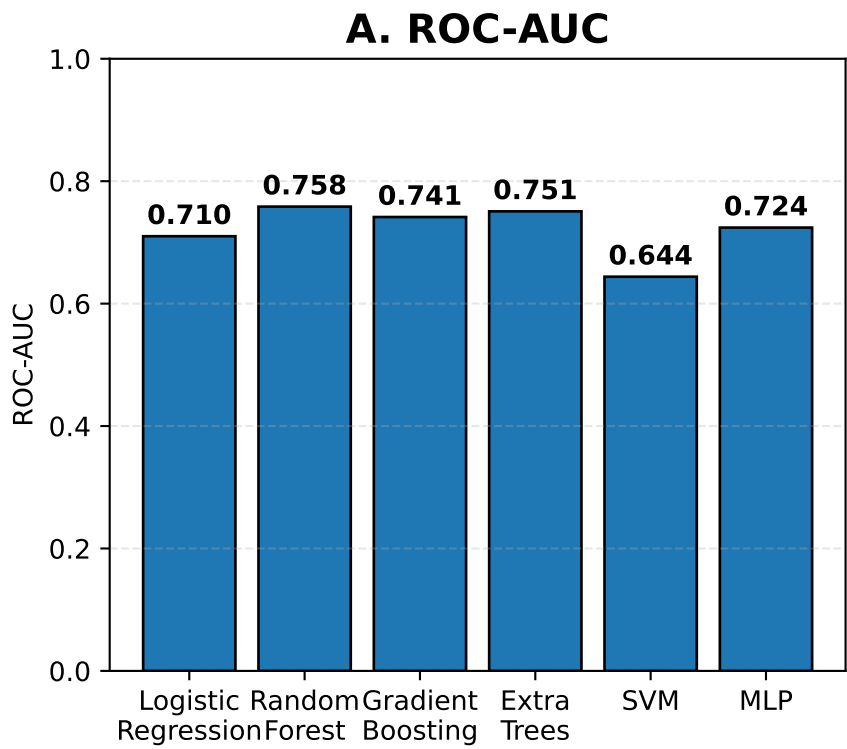


Strict global temporal validation of STX-LBD hypothesis prediction

Strict global temporal validation: pre-2016 KG → post-2015 recovery | Candidate pairs: 1,595 | Future positives: 309 (19.4%) | Future negatives: 1,286 (80.6%)



F. Summary metrics

Model	ROC	PR	RR	P@10	P@50	H@50	H@100
Logistic Regre	0.71	0.47	1.00	0.90	0.56	28	40
Random For	0.76	0.43	0.50	0.60	0.52	26	43
Gradient Boo	0.74	0.45	1.00	0.80	0.46	23	40
Extra Tree	0.75	0.43	1.00	0.60	0.52	26	43
SVM	0.64	0.41	1.00	0.70	0.58	29	41
MLP	0.72	0.38	0.50	0.60	0.40	20	39

Candidate pairs were generated from the pre-2016 dinoflagellate KG after hub filtering; positives were relationships recovered in the post-2015 KG.